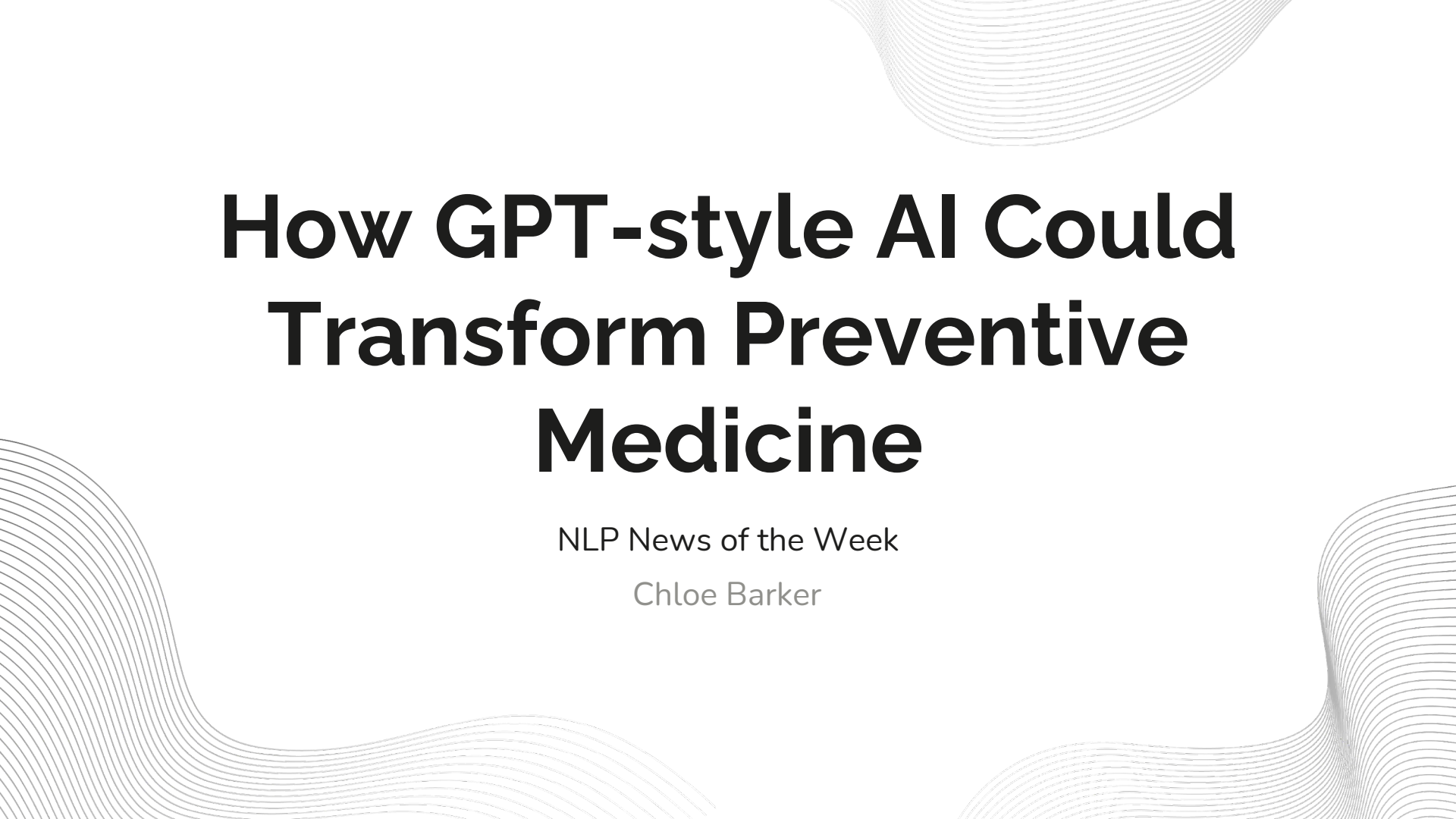


The background features decorative wavy lines in the corners, composed of many thin, concentric, curved lines that create a sense of depth and movement. These lines are light gray and frame the central text.

How GPT-style AI Could Transform Preventive Medicine

NLP News of the Week

Chloe Barker



The Problem

- Patients develop **multiple interacting conditions**, over time
- Most clinical risk tools predict one **disease at a time**
- They often **ignore order and timing** of diagnoses
- Risk is usually **detected after damage** begins

Key Gap: No unified way to model long-term health trajectories for various conditions



Meet *Delphi-2M*

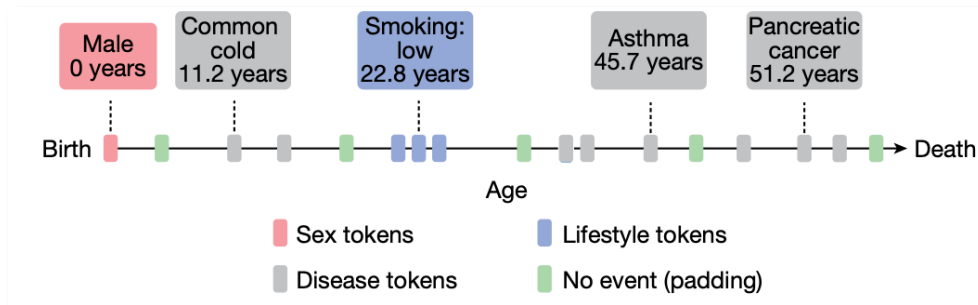
- **Predicts risk** for 1,000+ diseases simultaneously
- Forecasts disease **probability and timing** of events, up to 20 years ahead
- Built on a modified **GPT-style transformer**
- Treats health history as a **sequence**



How *Delphi-2M* Works

Health as Tokens Over Time

- Each diagnosis = a token (ICD-10 code)
- Tokens are ordered sequentially, by age
- Tokens Included:
 - Sex
 - Disease
 - Lifestyle (BMI, smoking, alcohol, etc.)
 - No event (padding)
- Model learns the grammar of disease progression



Sample Figure: Schematic of health trajectories based on ICD-10 diagnoses, lifestyle and healthy padding tokens, each recorded at a distinct age.

Credibility: Training, validation and testing data derived from:

- 400K UK Biobank participants
- 1.9M Danish disease registries

Limitations & Ethical Considerations

Limitations:

- Not ready for clinical deployment
- Predictions reflect probability, not certainty
- Performance varies by disease type
- Training data may introduce demographic bias

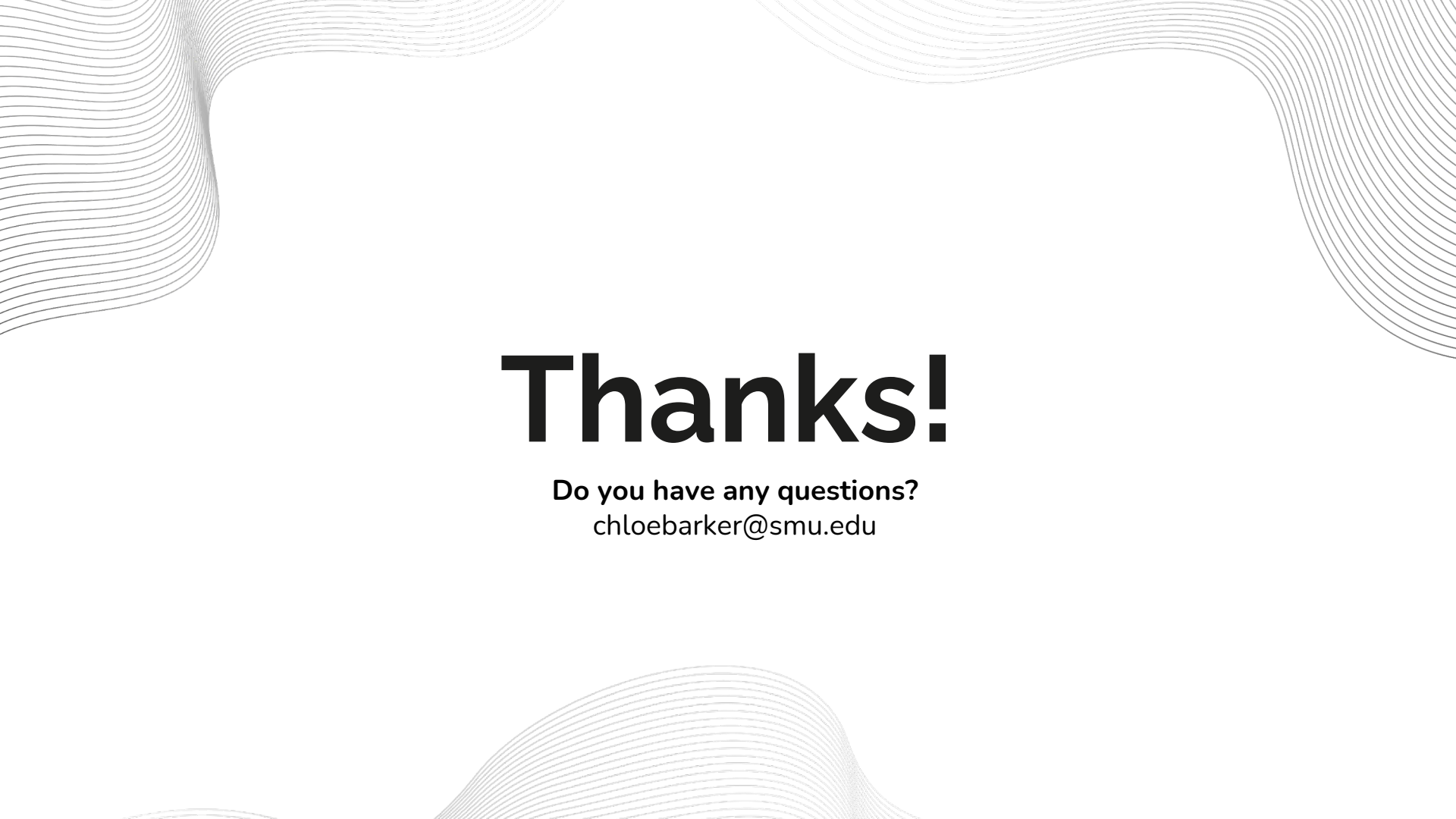
Ethical Considerations:

- Raises ethical questions about potential future risk
- Privacy and governance of sensitive health data

Why This Matters for Prevention

- Enables **earlier identification** of high-risk individuals
- Supports **preventive screening** before symptoms
- Allows population-level **forecasting**
- Shifts medicine from **reactive** → **preventive**
- Shows how NLP models can **predict health trajectories**
- Delphi-2M reframes healthcare as a **sequence modeling problem**
- Transformers can **learn the structure** of disease over time
- A major step toward **AI-driven** preventive medicine

Delphi-2M predicts not just **WHAT** is coming, but also **WHEN**

The background features decorative wavy lines in the corners, composed of many thin, concentric, curved lines that create a sense of depth and movement.

Thanks!

Do you have any questions?
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References

- Shmatko, A., Jung, A.W., Gaurav, K. et al. Learning the natural history of human disease with generative transformers. Nature 647, 248–256 (2025). <https://doi-org.proxy.libraries.smu.edu/10.1038/s41586-025-09529-3>